Class #16 - Inner Product Spaces

1. Def: An *inner product* on a vector space V is a function that, to each pair of vectors u and v in V, associates a _____number $\langle u, v\rangle$ and satisfies the following axioms, for all u, v, w in V and all scalars c:

   (a) $\langle u, v\rangle =$ _____

   (b) $\langle u + v, w\rangle =$ _____

   (c) $\langle cu, v\rangle =$ _____

   (d) $\langle u, u\rangle \geq$ _____ and $\langle u, u\rangle = 0$ if and only if _____.

   A vector space with an inner product is called an *inner product space.*

2. Def: Let V be an inner product space. Then

   (a) the *length* or *norm* of a vector v is the scalar _____,

   (b) the distance between u and v is _____, and

   (c) the vectors u and v are orthogonal if _____.

3. Example:

   (a) Let $t_0, \cdots, t_n$ be distinct real numbers. For p and q in $\mathbf{P_n}$, define

   $$\langle p, q\rangle = p(t_0)q(t_0) + p(t_1)q(t_1) + \cdots p(t_n)q(t_n).$$

...

(b) Let $V = \mathbf{P_4}$ with the inner product defined in (a) involving evaluation of polynomials at $-2, -1, 0, 1$, and 2.
Let $W = \mathbf{P_2}$. (Note that W is a _____of _____.)
Let $B = \{1, t, t^2\}$. (Then B is a _____of _____.) Is B an orthogonal set?

Produce an orthogonal basis for W by applying the Gram-Schmidt process to the basis B.

(c) Let $S = \{p_0, p_1, p_2\}$ be the orthogonal basis of W found in (b). Find the best approximation to $p(t) = 5 - \frac{1}{2}t^4$ by polynomials in W.

4. Example:

(a) For f, g in $C[a, b]$, set

$$\langle f, g \rangle = \int_a^b f(t)g(t)\,dt.$$

Show that this defines an inner product on $C[a, b]$.

...

(b) Let V be the space $C[0,1]$ with the inner product in (a), and let W be the subspace spanned by the polynomials $p_1(t) = 1, p_2(t) = 2t - 1$, and $p_3(t) = 12t^2$. Use the Gram-Schmidt process to find an orthogonal basis for W.

(c) Continuous functions are often approximated by linear combinations of sine and cosine functions. For instance, a continuous function might represent a sound wave, an electric signal of some type, or the movement of a vibrating mechanical system. It turns out that the set

$$\{1, \cos t, \cos 2t, \cdots, \cos nt, \sin t, \sin 2t, \cdots, \sin nt\}$$

forms an orthgonal set in $C[0, 2\pi]$.

Verify that $\cos mt$ and $\cos nt$ are orthogonal with respect to the inner product defined in (a) whenever m and n are positive integers with $m \neq n$.

(d) Let W be the subspace of $C[0, 2\pi]$ spanned by the orthogonal set in (c). Given any function $f \in C[0, 2\pi]$, the best approximation to f by functions in W is called the *n-th order Fourier approximation* to f on $[0, 2\pi]$, i.e, the best approximation $\hat{f} =$ _____. Find the nth-order Fourier approximation to the function $f(t) = t$ on the interval $[0, 2\pi]$.

5. The Cauchy-Schwarz Inequality:
   For all u, v in an inner product space V,

$$|\langle u, v\rangle| \leq ||u||||v||.$$

   Proof: